

Zhenrui Yue

Homepage | LinkedIn | GitHub | Google Scholar
yuezrhu@gmail.com | +1 650-605-5109 | Mountain View, CA 94043

Education

- University of Illinois Urbana-Champaign** Aug 2021 – May 2026
Ph.D. in Cognitive Science & Language Processing
• **Advisor:** Dr. Dong Wang; **Research:** Large Language Models, Reinforcement Learning & Information Retrieval
• **Thesis:** Advancing Language Models for Knowledge- and Reasoning-Intensive Tasks: From Training to Inference
- Technische Universität München** Apr 2019 – May 2021
M.Sc. in Robotics, Cognition, Intelligence
• **Thesis** (ETH Zurich w/ Dr. Stefan Feuerriegel): Question Answering under Domain Shift
• **Exchange** (UC San Diego w/ Dr. Julian McAuley): Supported by Max Weber Scholarship from 2019 to 2020
- Technische Universität München** Oct 2015 – Mar 2019
B.Sc. in Mechanical Engineering
• **Courses:** Mathematics, Mechanics, Control Theory, Simulation, Intro to CS, Software Eng., Algo & Data Struct. etc.

Experience

- Google DeepMind** Dec 2025 – Present
Research Scientist
• Improve Gemini post-training for open-ended generation, memory and p13n, delivering >5% benchmark gains
• Drive benchmark improvements for multiple models and agentic systems across reasoning, knowledge and ML tasks.
• Design autoresearch frameworks for self-improving post-training & modeling research (*Google AI Award recipient*).
- Meta MSL / GenAI** May 2025 – Dec 2025
Research Intern
• Worked with the safety post-training team on different reinforcement learning paradigms across diverse tasks
• Designed continuous space post-training methods for LLMs with enhanced reasoning performance and efficiency
• Introduced data-free self-improving training methods that autonomously evolve models without curated datasets
- Google DeepMind** May 2024 – Dec 2024
Student Researcher
• Explored the boundaries of long-context LLMs, designed various inference strategies and evaluated the outcome
• Introduced novel test-time scaling paradigms for long-context RAG with substantial gains on multiple benchmarks
- NVIDIA** May 2023 – Aug 2023
Applied Research Intern
• Investigated LLM-based recommendation methods and evaluated the performance of existing approaches
• Designed a two-stage LLM-based recommendation framework for efficient retrieval and personalized ranking

Selected Publications

Self-Instructing Policy Optimization: Unifying Reinforcement Learning with On-Policy Distillation

Zhenrui Yue, Huimin Zeng, Yueqi Wang, Yaokun Liu, Fengran Mo, Mung Yao Jia, Gyuseok Lee, Yang Zhang, Dong Wang
In Submission, 2026

Dr. Zero: Self-Evolving Search Agents without Training Data

Zhenrui Yue, Kartikeya Upasani, Xianjun Yang, Suyu Ge, Shaoliang Nie, Yuning Mao, Zhe Liu, Dong Wang
COLM 2026

FASA: Frequency-Aware Sparse Attention

Yifei Wang, Yueqi Wang, Zhenrui Yue, Huimin Zeng, Yong Wang, Ismini Lourentzou, Zhengzhong Tu, Xiangxiang Chu, Julian McAuley
ICLR 2026 [Paper] [Code]

Hybrid Latent Reasoning via Reinforcement Learning

Zhenrui Yue, Bowen Jin, Huimin Zeng, Honglei Zhuang, Zhen Qin, Jinsung Yoon, Lanyu Shang, Jiawei Han, Dong Wang
NeurIPS 2025 [Paper] [Code]

Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning

Bowen Jin, Hansi Zeng, **Zhenrui Yue**, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, Jiawei Han
COLM 2025 [Paper] [Code]

Inference Scaling for Long-Context Retrieval Augmented Generation

Zhenrui Yue*, Honglei Zhuang*, Aijun Bai, Kai Hui, Rolf Jagerman, Hansi Zeng, Zhen Qin, Dong Wang, Xuanhui Wang, Michael Bendersky
ICLR 2025 [Paper]

Retrieval Augmented Fact Verification by Synthesizing Contrastive Arguments

Zhenrui Yue, Huimin Zeng, Lanyu Shang, Yifan Liu, Yang Zhang, Dong Wang
ACL 2024 [Paper] [Code]

Linear Recurrent Units for Sequential Recommendation

Zhenrui Yue*, Yueqi Wang*, Zhankui He, Huimin Zeng, Julian McAuley, Dong Wang
WSDM 2024 [Paper] [Code]

LlamaRec: Two-Stage Recommendation using Large Language Models for Ranking

Zhenrui Yue, Sara Rabhi, Gabriel Moreira, Dong Wang, Even Oldridge
PGAI@CIKM 2023 [Paper] [Code]

Zero- and Few-Shot Event Detection via Prompt-Based Meta Learning

Zhenrui Yue, Huimin Zeng, Mengfei Lan, Heng Ji, Dong Wang
ACL 2023 [Paper] [Code]

Defending Substitution-Based Profile Pollution Attacks on Sequential Recommenders

Zhenrui Yue, Huimin Zeng, Ziyi Kou, Lanyu Shang, Dong Wang
RecSys 2022 [Paper] [Code]

Contrastive Domain Adaptation for Question Answering using Limited Text Corpora

Zhenrui Yue, Bernhard Kratzwald, Stefan Feuerriegel
EMNLP 2021 [Paper] [Code]

Talks

Google DeepMind: Automating the Continuous Improvement of LLMs	Jun 2026
NICE AI Talk: Dr. Zero: Self-Evolving Search Agents without Training Data	Feb 2026
Meta FAIR: Dr. Zero: Self-Evolving Search Agents without Training Data	Jan 2026
xAI: Post-Training LLMs with Multimodal & Continuous Representations	Sep 2025
Google DeepMind: Post-Training LLMs for Personalization and Reasoning	Aug 2025
Capital One: Hybrid Latent Reasoning via Reinforcement Learning	Jul 2025
Cohere: Inference Scaling for Long-Context Retrieval Augmented Generation	Jan 2025
LinkedIn: LLM for Efficient Sequential Recommendation	Aug 2024
Uber: Linear Recurrent Units for Recommendation	Jul 2024

Teaching

Computer Networks	Spring 2024
Concepts of Machine Learning	Fall 2023
Introduction to Database	Fall 2022

Awards

Shiyu Scholarship	2024
Bosch AIoT Scholarship	2021
Audi Scholarship	2018
Max Weber Program	2017

Miscellaneous

Coding: Python, C / C++ ; experiences with PyTorch, Jax, TensorFlow, Docker, Git, Bash, AWS, etc.
Languages: Native in Mandarin and Cantonese, proficient in English and German, beginner in Spanish
Service: AAAI, ARR (ACL, EMNLP, etc.), COLING, COLM, ICCV, ICLR, ICML, KDD, NeurIPS, SIGIR, WWW, etc.